

Welcome To Our Presentation

Presented by

H M Yeasin
ID: 20245103021
Md. Fahim Islam
ID: 20245103035
Md Sakibur Rahman
ID: 20245103091
Abu Ahmad Mahfuzor Rahman
ID:20245103092
MD.Abdur Rahim
ID: 20234103011

Presented To

Farha Akter Munmun

Lecturer

Department of Electrical and
Electronic Engineering

CapScan: What It Is & Why It Matters

What is CapScan?

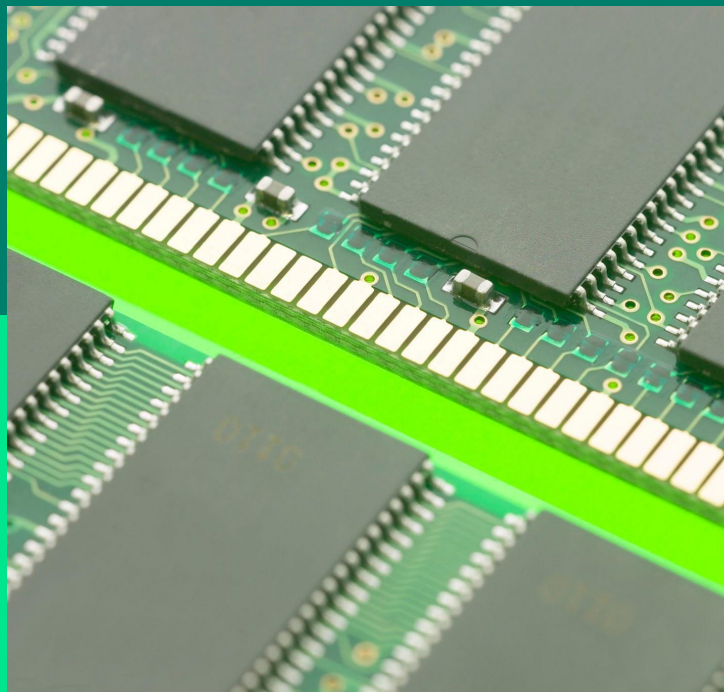
- Desktop tool to verify memory card & USB authenticity
- Checks real capacity, integrity, and performance

Why verification matters

- Protects data and ensures reliability
- Confirms true device capacity

Issues with fake/corrupt devices

- False capacity reporting
- Data loss & corruption risks
- Slow performance from low-quality components



Problem & Objectives

Problem Statement

- *Widespread fake or corrupted storage devices.
- *False capacity claims leading to data loss.
- *No simple tool for authenticity verification.

Project Objectives

- *Verify actual storage capacity.
- *Check device model authenticity.
- *Test data integrity & performance.
- *Provide clear, user-friendly scan results.



Tools & Technologies Used

- **Visual Studio Code (VS Code):** Development & debugging
- **Python 3:** Device detection, testing, GUI, database
- **Tkinter:** GUI design for scans and results
- **SQLite (capscan_db):** Stores scan history and results
- **Disk/Device Monitoring (smartctl):** Helps monitor storage health

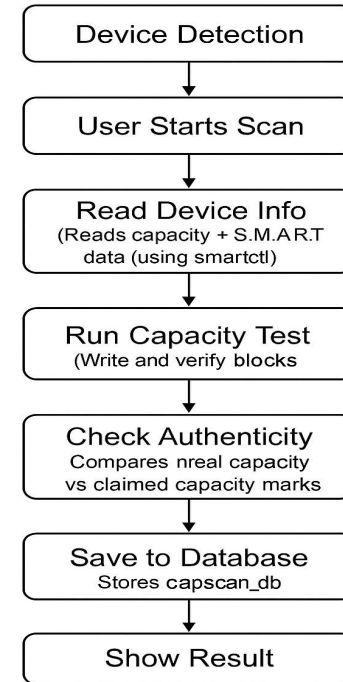
Scan History Table Schema:

Column Name	Type	Description
id	INTEGER	Unique scan record ID
device	TEXT	Device drive letter or identifier
serial	TEXT	Device serial number
claimed	REAL	Advertised capacity (MB/GB)
true	REAL	Actual verified capacity (MB/GB)
result	TEXT	Scan result (e.g., Genuine, Info)
speed	REAL	Read/Write speed (MB/s)
details	TEXT	Scan details and free space info
date	DATETIME	Date and time of the scan

Workflow

- Device Detection
- User Starts Scan
- Read Device Info
- Run Capacity Test
- Check Authenticity
- Save to Database
- Show Result
- Displays the result in the GUI.

CapScan – Simple Working Flow





Features Overview

#1

Detects fake or over-reported storage capacity to ensure users get genuine devices.

#2

Checks for corruption during read/write operations to protect data integrity.

#3

Measures device read/write performance for reliable speed assessment.

#4

Performs device health check to monitor overall storage condition.

#5

Stores records of previous scans for easy tracking and reference.

Project Results and Constraints

Results

THE PROBLEM WE'RE TACKLING

- Large-capacity testing can take time
- Hardware-dependent results
- Requires proper system permissions
- Health check not possible for small devices
- Only tested on Windows 10/11

- **Sample scan shows real vs reported capacity.**
- **Detects corrupted or fake devices**
- **Measures read/write speed for performance analysis**
- **Health check available for larger devices only**

Conclusion & Future Plans

Conclusion:

- CapScan helps verify real capacity and detect fake or corrupted storage devices
- Ensures data safety and reliable performance
- Provides user-friendly results and scan history

Future Enhancements:

- Cloud-based report syncing for access across devices
- Mobile app support for on-the-go scanning
- Expanded device database for more models
- Real-time model verification and updates
- Advanced predictive health monitoring



T H A N K

Y O U !